

A Proposal for a Master's Thesis

To be Entitled

PrEP Data Analysis and Reporting Tool (PrEP-DART)
A data Pipeline developed on Health Data Lab Frameworks
by

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1 Title

PrEP Data Analysis and Reporting Tool (DART): A Data Pipeline Developed on Health Data Lab Frameworks

2 Abstract

HIV pre-exposure prophylaxis (PrEP) programs are essential in preventing the transmission of HIV, requiring efficient monitoring and reporting frameworks to assess their reach and impact. This master thesis aims to develop a robust and scalable data analysis pipeline for reporting HIV-PrEP indicators, leveraging anonymized health data and aligning with the indicator framework defined by the European Centre for Disease Prevention and Control (ECDC) [1]. Using anonymized datasets provided by the Health Data Lab, the methodology automates data cleaning, integration, and transformation processes to compute and report key metrics. It integrates relational data across multiple tables, as defined by the dataset's entity-relationship model (ER Diagrams), and maps indicators to relevant fields, such as patient demographics, prEP prescriptions, adherence records, and laboratory results, for computational calculation [2]. The methodology outputs comprehensive reports, offering insights into program performance while remaining adaptable to future extensions or real-world data. This master thesis project aims to address the critical challenges in public health informatics, such as the automation of complex reporting tasks, by focusing on the development of a scalable data analysis pipeline. By reducing the time and effort required for manual reporting, the proposed pipeline not only supports efficient monitoring of HIV-PrEP programs but also serves as a blueprint for scalable and privacy-preserving data solutions in healthcare [3]. The results are expected to contribute to public health decision-making and improve resource allocation for HIV prevention strategies.

3 Introduction

Health Data: Resources and Tools from FDZ Gesundheit (DaTrav Structure)

The Health Data Lab (FDZ Gesundheit) is a research infrastructure that provides access to anonymized health-related datasets collected from healthcare providers, insurance claims, and public health institutions in Germany. These datasets are curated to support research in public health, epidemiology and health informatics while ensuring strict compliance with data protection regulations. The Health Data Lab (FDZ Gesundheit) provides descriptions of resources and documentation to support researchers in accessing and analyzing health-related datasets. FDZ Gesundheit offers a detailed description of the available datasets, outlining their structure, variables, and interrelationships. A dedicated platform includes comprehensive information about the datasets, such as variable definitions, coding systems, and data formats. To assist researchers in understanding the data architecture, the site provides **Entity-Relationship (ER) diagrams**, which visually map the relationships between tables, facilitating efficient data integration and analysis. Additionally, the main FDZ Gesundheit website explains the broader context and purpose of the data lab, emphasizing its role in offering secure access to health data for research. The site highlights the measures taken to ensure data anonymization, including techniques such as k-anonymity, pseudonym replacement, and selective sampling, which preserve privacy while maintaining data utility. This ensures that sensitive

health data can be analyzed ethically and responsibly. Furthermore, FDZ Gesundheit focuses on enabling data-driven public health research by providing secure and well-documented data pipelines for analysis. These combined resources empower researchers to work confidently with complex health datasets, allowing for robust analyses and informed decision-making. Together, the documentation and tools provided by FDZ Gesundheit serve as a foundational database for advancing public health research in compliance with privacy and ethical standards.

PrEP Indicators: Measuring the Impact of HIV Prevention Strategies

The European Centre for Disease Prevention and Control (ECDC) has established a comprehensive framework for monitoring HIV pre-exposure prophylaxis (PrEP) programs across the EU/EEA [6]. This framework includes indicators that are categorized into three domains: pre-uptake, uptake and coverage, and continued and effective use. Pre-uptake indicators evaluate factors influencing PrEP adoption, such as service availability, awareness among potential users, and willingness to use prEP. Uptake and coverage indicators measure the actual utilization and reach of PrEP, focusing on metrics like the number of current and new PrEP users, overall PrEP coverage, and the PrEP-to-Need Ratio (PnR), which serves as a proxy for assessing coverage. Continued and effective use indicators examine the sustainability and effectiveness of PrEP over time, tracking metrics like recent PrEP use among individuals newly diagnosed with HIV and the continuation rates of users after 12 months. These indicators are further classified into core, supplementary, and optional categories, based on their importance and feasibility for regular monitoring. Core indicators are essential and feasible for routine use, supplementary indicators provide valuable insights but may require additional resources, and optional indicators offer deeper understanding but are less practical for standard reporting. By implementing this structured approach, PrEP program performance can be systematically assessed, gaps identified, and strategies optimized to enhance HIV prevention efforts across diverse populations. This robust monitoring framework enables data-driven decision-making to advance public health objectives. Monitoring will be vital to understand whether programs succeed in engaging people into using PrEP appropriately, and in reducing the HIV epidemic. [5]

PrEP Surv: A Comprehensive Initiative for Monitoring HIV Prevention in Germany

The project “Surveillance of HIV pre-exposure prophylaxis provision within the SHI system in Germany (PrEP Surv), initiated by Robert Koch-Institute and financed by the German Federal Ministry of Health (BMG), represents a pivotal effort in the surveillance and evaluation of HIV pre-exposure prophylaxis (PrEP) among HIV-negative populations in Germany. Building on the outcomes of the prior project, "Evaluation of the Introduction of HIV Pre-Exposure Prophylaxis as a Benefit of Statutory Health Insurance (EvE-PrEP)", PrEP Surv was designed to monitor the efficacy and reach of PrEP within the framework of statutory health insurance (SHI). The initiative encompasses key components of PrEP care, including the administration of [tenofovir disoproxil + emtricitabine, TDF/FTC] [4], patient counseling, medication supply, and associated medical precautions such as diagnostic follow-ups. The launch of PrEP Surv marked the translation of evaluation findings into a structured surveillance program, aligned with EU-PrEP monitoring guidelines. A primary focus of the initiative was to assess the incidence of HIV and sexually transmitted infections (STIs) among the prEP users in Germany. Project prEP Surv operated across three interconnected modules: (1) secondary data collection from pharmacies, health insurers, and FDZ, (2) enhancement of data collection from German HIV specialty care centers within the dāgna network to

strengthen PrEP service reporting, and (3) collaboration with communities with a potential need for PrEP to identify barriers to PrEP access and address unmet needs in HIV prevention. By leveraging these comprehensive data streams, PrEP Surv aimed to advance the penetration of PrEP services and establish an evidence-based foundation for future HIV monitoring strategies in Germany.

4 Objectives

1. To develop a robust data pipeline, named PrEP-DART(prEP- Data Analysis and Reporting Tool), for automating data processing and reporting.
2. To map ECDC-defined PrEP indicators to the Health Data Lab datasets for computational calculation. (Core, supplementary and optional PrEP indicators, such as PrEP coverage, new PrEP Users, and PrEP-to-Need Ratio (PnR), in alignment with EU guidelines and ensuring the reporting system is adaptable for real-world data and future extensions.
3. Validation of HIR pipeline's performance using mock datasets from FDZ, ensuring accuracy and efficiency in indicator computation.

5 Literature Review

1. A Review of Anonymization for Healthcare Data [7]
2. Monitoring HIV pre-exposure prophylaxis programs in the EU/EEA (ECDC, 2024)
3. Algorithms to anonymize structured medical and healthcare data: A systematic review [8]
4. A richly interactive exploratory data analysis and visualization tool using electronic medical records [9]
5. Effective Environmental Public Health Surveillance Programs: A Framework for identifying and evaluating data resources and indicators [10]
6. Interactive Visualization Applications in Population Health and Health Services Research: Systematic Scoping Review [11]

6 Methodology

6.1 Data Collection

The master thesis project utilizes synthetically generated datasets to simulate real-world data while preserving privacy and confidentiality requirements. The synthetic data is created using the framework provided by the Health Data Lab (FDZ Gesundheit). The Public Use File (PUF) repository enables researchers to develop and test analysis pipelines with datasets that mimic the structure and schema of FDZ's actual datasets, including entity-relationship (ER) models and variable specifications. The PUF data is modeled to replicate the relationships between various entities in the FDZ datasets. These include patient demographics, prescription data, healthcare service utilization and regional health statistics. The data structure is based on ER diagrams that define the relationship between the tables, making it suitable for developing and testing relational databases. The synthetic nature of PUF data ensures privacy by destroying correlations, effectively preventing the extraction of sensitive insights or interferences that could compromise confidentiality. While the dataset will be structured like real-world data, it does not maintain the underlying relationship or patterns that exists in the actual data. This complementary nature of PUF data helps researchers focus on building robust analytical solutions without compromising privacy or requiring access to sensitive information.

6.2. Pipeline Architecture: *prEP-DART* (*prEP- Data Analysis and Reporting Tool*)

The development of the *prEP-DART* (*prEP- Data Analysis and Reporting Tool*) pipeline involves creating a robust, automated system for processing and analyzing health datasets to compute key PrEP indicators. This process includes multiple stages, each designed to ensure the pipeline is scalable, efficient, and compliant with privacy standards. The pipeline begins with dataset model provided by the Health Data Lab (FDZ Gesundheit), which serves as the foundation for the PUF generated data. Further, mapping of relevant fields in the generated data to the ECDC-defined indicators for accurate calculations. The Public Use File (PUFs) serve as an intermediary dataset, allowing the pipeline to operate on secure and anonymized information without compromising analytical utility. At the core of the architecture is the *prEP- Data Analysis and Reporting Tool* pipeline, which automates: (1) Data cleaning and transformation of Public Use File (PUF) datasets (2) Computational calculations for key indicators like prEP coverage, uptake, and the prEP-to-Need Ratio (PnR) (3) Validation and testing with different mock PUF datasets. The pipeline outputs **tabular summaries** and **visualizations**, created using Python libraries such as Matplotlib, Seaborn, and Plotly. These outputs are not designed for real-world public health monitoring but rather serve as benchmarks for demonstrating the pipeline's functionality and readiness for future integration with real-world datasets. This architecture demonstrates how the *prEP- Data Analysis and Reporting Tool* pipeline integrates with PUF datasets, aligns them with standardized PrEP indicators, and transforms anonymized data into actionable insights. By leveraging privacy-preserving methods and automated workflows, the pipeline lays the foundation for efficient and ethical reporting and validation to support public health monitoring and improve resource allocation for HIV prevention strategies.

6.3 Data Analysis

The project will assess its impact through key metrics aligned with the ECDC-defined indicators, such as PrEP adoption rates, coverage, and the PrEP-to-Need Ratio (PnR). Overall the PUF structure maintains the same structure as the DaTrav structure with core components of Entity-Relationship (ER) models, tables and variables, Data models and structural components such as Primary Keys (PK), relationships, Hierarchical Variables, Anonymized data and standardization. The data pre-processing steps include handling missing or inconsistent values, standardizing formats across datasets (date formats, variable names) followed by removing duplicates and irrelevant records. Secondly, the algorithm uses ER diagrams to integrate relational datasets, ensuring consistency in linking tables such as patient records, prescriptions, and service data. Further the algorithm maps relevant data fields to the core, supplementary and optional indicators defined by the European Centre for Disease Prevention and Control (ECDC). These computations are performed for specific timeframes (eg., annually and quarterly) to monitor trends. The validations will be performed using running test cases on mock datasets to ensure accuracy and comparing calculated metrics against benchmark values and historical data. The data analysis will be distinguished as (1) Descriptive Analysis, to provide an overview of the dataset and key metrics (2) prEP-Trend analysis, to monitor changes in prEP indicators over time (3) Comparative Analysis, to identify disparities in prEP adoption across demographics or regions using group-wise comparisons (e.g., by age, gender, geography)

(4) Predictive Insights, to forecast future trends in prEP adoption and its impact on historical data (5) Indicator-Specific Insights, to identify gaps in service provisions(prEP-coverage), to highlight the effectiveness of the program in reducing new infections(PnR) and to evaluate adherence and retention among prEP users(continuation rates). The resulting insights empower beneficiaries to make data-driven decisions, optimize resource allocation, and enhance the overall impact of HIV prevention programs. This pipeline serves as a validation-blueprint for scalable and privacy-compliant data solutions in public health informatics in the future.

7 Timeline

Phase	Duration	Deliverables
Literature Review	3 weeks	Annotated bibliography, draft background on PrEP indicators, privacy techniques
Data Collection	2 weeks	Validation of FDZ Dataset Integration and PUF Generation
Framework Development	2 months	A functional prototype of the data pipeline capable of processing synthetic PUF datasets and computing key PrEP indicators.
Indicator Mapping	3 weeks	Mapping of ECDC-defined indicators to the PUF datasets
Data Analysis	1 month	Statistical models, calculations of PrEP indicators, and initial visualizations
Automated Reporting	2 weeks	Reports with tables and visualizations, final testing
Validation and Testing	2 weeks	Validation with different mock datasets, accuracy testing
Thesis Writing	1 month	Final thesis including methodology, results, and discussions

8 References

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